SRM TRP Engineering College

(An AUTONOMOUS Institution)

Tiruchirappalli

Higher Order Assessment Instructions with Rubrics

Department	Computer Science and Engineering							
Programme:	B.E.		Semester:	I	AY:	2026 – 2027 [ODD]		
Faculty Handling:	Mr. A. Sendhooran, AP/CSE							
Course Code & Title:	2611CSSJ102 & Web Essentials							
Assessment Task:	Mini Project (Project Based Learning)		Assessment Type:	Term Work / Individual		Required Hrs:	2 Weeks	
Assessment Question based on:	Open Ended	✓	Problem Based Learning			Creative Thinking	✓	
	Close Ended		Project Based Learning	✓		Critical Thinking	✓	
	Any Other Type	MCQ or any other	Challenge Based Learning			Level of Thinking (RBT)	K3/K6	
SDG to be Addressed:	SDG is Required	Yes	SDG Number (Specify or Mention Students Choice)	SDG 4 / 9		Target Number (If Applicable)	4.4 / 9.c	
SDG Integration Level: (If Addressed)	Level 1: Awareness		Level 2: Design Alignment	✓		Level 3: Impact and Measurement		

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RIASEC Context	Realistic ✓	Investigative ✓	Artistic ✓	Social	Enterprising	Conventional
Presentation Format:	Live Demonstration (localhost) / PDF Report / GitHub Repository					
Max Marks:	40 Marks					
Relevant Course Outcome(s):	CO1, CO2					
Submission Deadline:	09/10/2026					
Student Name & Register No.	Bavadharani L – TRP26CS34 (I CSE A)					
Allocated Website Topic	Department of Robotics and Automation (sample topic)					

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1. Topic and Aim

Item	Details
Course	2611CSSJ102 – Web Essentials (Units 1 and 2)
Topic	Department of Robotics and Automation – informational website
Student	Bavadharani L, TRP26CS34, I CSE A
Aim	To plan, build and test a multi-page HTML5 website for a college department that gives prospective students information about the department and lets them register for an open-house visit, and to run it through a local Web server.

Objectives

- Create six connected pages (Home, About, Courses, Register, Contact and Unit 1 Evidence) with valid, semantic HTML5.
- Use tables, lists, an accessible form with HTML5 input types, images, and video with captions.
- Serve the site on localhost and verify it with browser Developer Tools and the W3C validator.
- Explain the browser–DNS–server process, URL structure, hosting choice and security precautions.

2. Components Used

2.1 Hardware and Software

Component	Specification / Purpose
Computer	Any PC or laptop, 4 GB RAM, Windows / Linux / macOS
Code editor	Visual Studio Code – writing HTML and CSS
Web browser	Google Chrome – viewing pages and Developer Tools (Elements, Network, Console)
Local Web server	Python http.server on port 8000 (Apache in XAMPP can also be used)
Validator	W3C Nu HTML Checker – checks the HTML structure
Version control	GitHub repository – submission of source code

2.2 HTML5 Components

Area	Elements / attributes used	Where used
Document structure	<!DOCTYPE html>, html lang, head, meta charset, meta viewport, title, link	All pages
Semantic layout	header, nav, main, section, article, aside, footer, figure, figcaption, address, time	All pages
Text	h1–h3, p, strong, em, mark, small, sup, sub, q, blockquote, pre, code	about.html, index.html
Lists	ul, ol, dl / dt / dd	index.html, about.html
Links	Relative, absolute, fragment (#office-hours), mailto:, tel:, download, target="_blank" with rel="noopener"	Navigation, contact.html
Tables	table, caption, thead, tbody, tr, th (scope col / row / colgroup), td, rowspan, colspan	courses.html, contact.html
Forms	form (action, method="post"), fieldset, legend, label, input, select, option, textarea, button	register.html
Input types	text, email, tel, date, number, radio, checkbox, file	register.html

Area	Elements / attributes used	Where used
Validation	required, minlength, maxlength, pattern, min, max, accept	register.html
Media	img with alt, video with controls, two source formats, track (captions), poster	index.html, about.html
Accessibility	aria-current="page", aria-label, aria-describedby, lang="ta"	All pages

2.3 Project Folder

File / Folder	Contents
index.html	Home: introduction, highlights, call to action, tour video
about.html	History, mission, facilities, contact summary
courses.html	Accessible course table with rowspan and colspan
register.html	Open-house registration form
contact.html	Address, email, telephone, download link, office hours
evidence.html	Unit 1 evidence: request–response, URL analysis, hosting, security
style.css	Basic readability styles
images/, media/, documents/	Banner and lab images, tour video (WebM, MP4) with captions, prospectus PDF
README.txt	Project details and steps to run

2.4 Procedure

1. Prepared the requirement analysis and a site map of the six pages.
2. Created the department-site folder with images, media, documents and screenshots sub-folders.
3. Wrote each page in VS Code using the same header, navigation and footer.
4. Started the local server with "python -m http.server 8000" and opened http://localhost:8000.

5. Checked every page with Developer Tools and the W3C validator, recorded tests T01–T12 and fixed the defects found.
6. Uploaded the project folder to GitHub and submitted the repository link.

3. Program

The complete source code of the six pages and the stylesheet is given below.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="Home page of the Department of Robotics and Automation.">
 <title>Home | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html" aria-current="page">Home</a></li>
     <li><a href="about.html">About</a></li>
     <li><a href="courses.html">Courses</a></li>
     <li><a href="register.html">Register</a></li>
     <li><a href="contact.html">Contact</a></li>
     <li><a href="evidence.html">Unit 1 Evidence</a></li>
   </ul>
  </nav>
```

```
 </header>

 <main id="main-content">
  <!-- Home page: introduction, highlights and call to action -->
  <section id="welcome">
    <h2>Learn. Build. Make machines think responsibly.</h2>
    <figure>
      <img src="images/robotics-banner.png" alt="Department banner showing a robotic control system with input, hidden and output layers"
width="1200" height="450">
      <figcaption>Figure 1: A robotic control system, the core idea behind many of our courses.</figcaption>
    </figure>
    <p>The Department of Robotics and Automation prepares students to design <strong>automated robotic systems</strong> that solve real
problems in health, agriculture, transport and education. Our programme combines strong mathematics, programming and <em>hands-on
laboratory work</em>.</p>
  </section>

  <section id="highlights">
    <h2>Highlights</h2>
    <ul class="highlight-list">
      <li>Four-year B.E. programme with an intake of 60 students</li>
      <li>Robotics Laboratory with 40 GPU-enabled workstations</li>
      <li>Industry-led workshops every semester</li>
      <li>Robotics Club and an annual project expo</li>
    </ul>
  </section>

  <section id="explore">
```

```
    <h2>Explore the department</h2>
    <ol>
      <li><a href="about.html">Read about our history, mission and facilities</a></li>
      <li><a href="courses.html">View the course and lab sessions table</a></li>
      <li><a href="contact.html">Find our address and office hours</a></li>
    </ol>
    <p><a href="register.html"><strong>Register for the open-house visit</strong></a> to meet faculty and see the laboratories.</p>
  </section>

  <aside>
    <h2>Department tour</h2>
    <video controls width="640" height="360" preload="metadata" poster="images/tour-poster.png">
      <source src="media/department-tour.webm" type="video/webm">
      <source src="media/department-tour.mp4" type="video/mp4">
      <track src="media/department-tour.vtt" kind="captions" srclang="en" label="English" default>
      Your browser does not support the video element. Read the <a href="media/department-tour.vtt">tour transcript</a>.
    </video>
    <p class="hint">The video has English captions. A text transcript is also available.</p>
  </aside>
 </main>

 <footer>
  <p><small>&copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample
project for 2611CSSJ102 Web Essentials.</small></p>
  <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
```

```
</html>
```

about.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="History, mission, facilities and contact summary of the Department of Robotics and Automation.">
 <title>About | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html">Home</a></li>
     <li><a href="about.html" aria-current="page">About</a></li>
     <li><a href="courses.html">Courses</a></li>
     <li><a href="register.html">Register</a></li>
     <li><a href="contact.html">Contact</a></li>
     <li><a href="evidence.html">Unit 1 Evidence</a></li>
   </ul>
  </nav>
```

```
 </header>

 <main id="main-content">
  <article id="history">
   <h2>Our History</h2>
   <p>The department started in <time datetime="2020-07-01">July 2020</time> with 30 students and two laboratories. Today it supports
<strong>240 students</strong> and a team of 16 faculty members.</p>
   <blockquote cite="https://department.example.edu/about.html">
    <p>Good robotics begins with careful design and safe control.</p>
   </blockquote>
   <p>&mdash; Head of the Department, during the first convocation.</p>
  </article>

  <section id="mission">
   <h2>Mission and Objectives</h2>
   <ul>
    <li>Teach the mathematics behind learning algorithms, such as minimising a loss <var>L</var> = (<var>y</var> &minus;
<var>&#375;</var>)<sup>2</sup>.</li>
    <li>Train students to build and evaluate models on real datasets.</li>
    <li>Promote ethical and <mark>responsible use of Robotics</mark>.</li>
   </ul>
   <p>As our students often say, <q>a robot is only as safe as its control system</q>.</p>
  </section>

  <section id="facilities">
   <h2>Facilities</h2>
   <figure>
```

```
      <img src="images/plc-lab.png" alt="Illustration of the Robotics Laboratory with four workstations displaying bar charts" width="800" height="450">
      <figcaption>Figure 2: The Robotics Laboratory.</figcaption>
    </figure>
    <dl>
      <dt>Robotics Laboratory</dt>
      <dd>40 workstations with GPUs for deep learning practice.</dd>
      <dt>PLC Training Laboratory</dt>
      <dd>Cameras, robot kits and sensors for computer-vision projects.</dd>
      <dt>Department Library</dt>
      <dd>Over 1,200 books and journals on robotics and automation, including CO<sub>2</sub> emission prediction case studies.</dd>
    </dl>
    <h3>Sample code shown in our first lab</h3>
    <pre><code>model = train(data, epochs=10)
print(model.accuracy)</code></pre>
  </section>

  <section id="contact-summary">
    <h2>Contact Summary</h2>
    <p>Office: Block C, First Floor. Email <a href="mailto:robotics@department.example.edu">robotics@department.example.edu</a>. Full details are on the <a href="contact.html">contact page</a>.</p>
  </section>
 </main>

 <footer>
  <p><small>&amp;copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample project for 2611CSSJ102 Web Essentials.</small></p>
```

```
  <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
</html>
```

courses.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="Courses and laboratory sessions offered by the Department of Robotics and Automation.">
 <title>Courses | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html">Home</a></li>
     <li><a href="about.html">About</a></li>
     <li><a href="courses.html" aria-current="page">Courses</a></li>
     <li><a href="register.html">Register</a></li>
     <li><a href="contact.html">Contact</a></li>
```

```
      <li><a href="evidence.html">Unit 1 Evidence</a></li>
    </ul>
  </nav>
 </header>

 <main id="main-content">
  <section id="course-table">
   <h2>Courses and Laboratory Sessions</h2>
   <p>The table lists first-year courses offered by the department. Each row is one course; the first column names the course, and the following columns give its semester, credits and weekly hours. Laboratory courses have separate theory and practical hours.</p>
   <div class="table-wrap">
    <table>
     <caption>Table 1: First-year courses, credits and weekly hours</caption>
     <thead>
      <tr>
       <th scope="col" rowspan="2">Course</th>
       <th scope="col" rowspan="2">Semester</th>
       <th scope="col" rowspan="2">Credits</th>
       <th scope="colgroup" colspan="2">Hours per week</th>
      </tr>
      <tr>
       <th scope="col">Theory</th>
       <th scope="col">Practical</th>
      </tr>
     </thead>
     <tbody>
      <tr>
```

```
 <th scope="row">Python Programming</th>
 <td>I</td>
 <td>4</td>
 <td>3</td>
 <td>2</td>
</tr>
<tr>
 <th scope="row">Web Essentials</th>
 <td>I</td>
 <td>3</td>
 <td>2</td>
 <td>2</td>
</tr>
<tr>
 <th scope="row">Robotics Mathematics</th>
 <td>II</td>
 <td>4</td>
 <td>4</td>
 <td>0</td>
</tr>
<tr>
 <th scope="row">Industrial Robotics</th>
 <td>II</td>
 <td>4</td>
 <td>3</td>
 <td>2</td>
</tr>
```

```
        <tr>
          <th scope="row">PLC Automation Seminar</th>
          <td>II</td>
          <td>1</td>
          <td colspan="2">Weekly two-hour seminar</td>
        </tr>
       </tbody>
      </table>
    </div>
    <p class="hint">Credits are earned on passing the end-semester examination. Hours may change slightly each year.</p>
  </section>
 </main>

 <footer>
   <p><small>&copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample
project for 2611CSSJ102 Web Essentials.</small></p>
   <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
</html>
```

register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
```

```
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="Open-house visit registration form for the Department of Robotics and Automation.">
 <title>Register | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html">Home</a></li>
     <li><a href="about.html">About</a></li>
     <li><a href="courses.html">Courses</a></li>
     <li><a href="register.html" aria-current="page">Register</a></li>
     <li><a href="contact.html">Contact</a></li>
     <li><a href="evidence.html">Unit 1 Evidence</a></li>
   </ul>
  </nav>
 </header>

 <main id="main-content">
  <section id="registration">
   <h2>Open-House Visit Registration</h2>
   <p>Fill in this form to book a place at the department open house. Fields marked <strong>required</strong> must be completed.</p>
   <!-- The form uses POST so personal details are not shown in the URL. Server-side validation is still needed when this is hosted. -->
   <form action="https://department.example.edu/register" method="post" enctype="multipart/form-data">
```

```
<fieldset>
  <legend>Personal details</legend>
  <label for="full-name">Full name (required, 3&amp;ndash;60 characters)</label>
  <input type="text" id="full-name" name="full_name" required minlength="3" maxlength="60" autocomplete="name">

  <label for="email">Email address (required)</label>
  <input type="email" id="email" name="email" required autocomplete="email">

  <label for="phone">Mobile number</label>
  <input type="tel" id="phone" name="phone" pattern="[0-9]{10}" aria-describedby="phone-hint" autocomplete="tel">
  <span class="hint" id="phone-hint">Format: 10 digits, for example 9876543210.</span>

  <label for="visit-date">Preferred visit date</label>
  <input type="date" id="visit-date" name="visit_date" min="2026-11-01" max="2026-11-30" required>

  <label for="visitors">Number of visitors (1&amp;ndash;4)</label>
  <input type="number" id="visitors" name="visitors" min="1" max="4" value="1" required>
</fieldset>

<fieldset class="choice">
  <legend>Mode of visit</legend>
  <input type="radio" id="mode-campus" name="visit_mode" value="campus" checked>
  <label for="mode-campus">On campus</label>
  <input type="radio" id="mode-online" name="visit_mode" value="online">
  <label for="mode-online">Online session</label>
</fieldset>
```

```
<fieldset class="choice">
  <legend>Areas of interest</legend>
  <input type="checkbox" id="int-ml" name="interests" value="machine-learning">
  <label for="int-ml">Machine learning</label>
  <input type="checkbox" id="int-vision" name="interests" value="computer-vision">
  <label for="int-vision">Computer vision</label>
  <input type="checkbox" id="int-robotics" name="interests" value="robotics">
  <label for="int-robotics">Robotics</label>
</fieldset>

<fieldset>
  <legend>Session and questions</legend>
  <label for="session">Session</label>
  <select id="session" name="session" required>
    <option value="">Choose a session</option>
    <option value="morning">Morning, 10:00 to 12:00</option>
    <option value="afternoon">Afternoon, 14:00 to 16:00</option>
  </select>

  <label for="question">Question for the faculty</label>
  <textarea id="question" name="question" rows="4" cols="40" maxlength="300"></textarea>

  <label for="marksheet">Upload Class XII mark sheet (optional)</label>
  <input type="file" id="marksheet" name="marksheet" accept=".pdf,.jpg,.png" aria-describedby="file-hint">
  <span class="hint" id="file-hint">PDF, JPG or PNG, up to 2 MB.</span>
</fieldset>
```

```
      <button type="submit">Submit registration</button>
    </form>
  </section>
 </main>

 <footer>
  <p><small>&copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample project for 2611CSSJ102 Web Essentials.</small></p>
  <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
</html>
```

contact.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="Address, email, telephone and office hours of the Department of Robotics and Automation.">
 <title>Contact | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
```

```
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html">Home</a></li>
     <li><a href="about.html">About</a></li>
     <li><a href="courses.html">Courses</a></li>
     <li><a href="register.html">Register</a></li>
     <li><a href="contact.html" aria-current="page">Contact</a></li>
     <li><a href="evidence.html">Unit 1 Evidence</a></li>
   </ul>
  </nav>
</header>

<main id="main-content">
  <section id="contact-details">
   <h2>Contact Us</h2>
   <p><a href="#office-hours">Jump to office hours</a></p>
   <address>
    Department of Robotics and Automation<br>
    SRM TRP Engineering College, Block C<br>
    Tiruchirappalli, Tamil Nadu 620 000
   </address>
   <ul>
    <li>Email: <a href="mailto:robotics@department.example.edu">robotics@department.example.edu</a></li>
    <li>Telephone: <a href="tel:+914310000000">+91 431 000 0000</a></li>
    <li>Prospectus: <a href="documents/robotics-prospectus-2026.pdf" download>Download the 2026 prospectus (PDF, 2 KB)</a></li>
```

```
      <li>Map: <a href="https://www.openstreetmap.org/#map=13/10.8050/78.6856" target="_blank" rel="noopener">View Tiruchirappalli on OpenStreetMap (opens in a new tab)</a></li>
    </ul>
    <p class="hint">The external link uses <code>rel="noopener"</code> so the new tab cannot control this page through <code>window.opener</code>.</p>
  </section>

  <section id="office-hours">
    <h2>Office Hours</h2>
    <table>
      <caption>Table 2: Department office hours</caption>
      <thead>
        <tr>
          <th scope="col">Day</th>
          <th scope="col">Hours</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <th scope="row">Monday to Friday</th>
          <td><time datetime="09:00">9:00</time> to <time datetime="17:00">17:00</time></td>
        </tr>
        <tr>
          <th scope="row">Saturday</th>
          <td><time datetime="09:00">9:00</time> to <time datetime="13:00">13:00</time></td>
        </tr>
        <tr>
```

```
        <th scope="row">Sunday</th>
        <td>Closed</td>
       </tr>
      </tbody>
    </table>
  </section>
 </main>

 <footer>
  <p><small>&copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample project for 2611CSSJ102 Web Essentials.</small></p>
  <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
</html>
```

evidence.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <meta name="description" content="Unit 1 evidence: request-response process, URL analysis, tools, hosting and security.">
 <title>Unit 1 Evidence | Department of Robotics and Automation</title>
 <link rel="stylesheet" href="style.css">
</head>
```

```
<body>
 <header>
  <h1>Department of Robotics and Automation</h1>
  <p>SRM TRP Engineering College, Tiruchirappalli</p>
  <nav aria-label="Primary">
   <ul>
     <li><a href="index.html">Home</a></li>
     <li><a href="about.html">About</a></li>
     <li><a href="courses.html">Courses</a></li>
     <li><a href="register.html">Register</a></li>
     <li><a href="contact.html">Contact</a></li>
     <li><a href="evidence.html" aria-current="page">Unit 1 Evidence</a></li>
   </ul>
  </nav>
 </header>

 <main id="main-content">
  <section id="request-response">
   <h2>What happens when a user opens this website</h2>
   <figure>
    <img src="images/request-response.png" alt="Diagram: browser sends URL to DNS, DNS returns IP address, browser sends HTTPS request
to web server, server returns HTML and media, browser renders the page" width="1000" height="300">
    <figcaption>Figure 3: Request&ndash;response process for this website.</figcaption>
   </figure>
   <ol>
    <li>The user types the <strong>URL</strong> into the <strong>browser</strong> (the client) and presses Enter.</li>
```

```
      <li>The browser asks a <strong>DNS</strong> resolver for the <strong>IP address</strong> of
<code>department.example.edu</code>.</li>
      <li>Using that IP address, the browser sends an <strong>HTTPS request</strong> (GET /index.html) to the <strong>Web
server</strong>.</li>
      <li>The server finds the file and sends a <strong>response</strong>: status code 200 and the HTML document.</li>
      <li>While reading the HTML, the browser sends further requests for the CSS, images and video.</li>
      <li>The browser <strong>renders</strong> the page from the HTML, CSS and media files.</li>
    </ol>
  </section>

  <section id="url-analysis">
    <h2>URL Analysis</h2>
    <p>Example URL: <code>https://department.example.edu:443/courses.html?level=ug#course-table</code></p>
    <table>
      <caption>Table 3: Parts of the example URL</caption>
      <thead>
        <tr><th scope="col">Part</th><th scope="col">Value</th><th scope="col">Meaning</th></tr>
      </thead>
      <tbody>
        <tr><th scope="row">Scheme</th><td>https</td><td>Secure HTTP protocol</td></tr>
        <tr><th scope="row">Domain name</th><td>department.example.edu</td><td>Name resolved to an IP address by DNS</td></tr>
        <tr><th scope="row">Port</th><td>443</td><td>Default HTTPS port (optional)</td></tr>
        <tr><th scope="row">Path</th><td>/courses.html</td><td>File requested on the server</td></tr>
        <tr><th scope="row">Query</th><td>?level=ug</td><td>Extra data sent to the server</td></tr>
        <tr><th scope="row">Fragment</th><td>#course-table</td><td>Section of the page to scroll to</td></tr>
      </tbody>
    </table>
```

```
    <h3>Relative paths</h3>
    <ul>
      <li><code>images/plc-lab.png</code> in about.html: the browser joins it with the current folder, giving
<code>http://localhost:8000/images/plc-lab.png</code>.</li>
      <li><code>contact.html#office-hours</code>: opens contact.html in the same folder and scrolls to the element with <code>id="office-
hours"</code>.</li>
    </ul>
  </section>

  <section id="environment">
    <h2>Development Environment</h2>
    <p>Editor: Visual Studio Code. Browser: Google Chrome. Local server: Python <code>http.server</code> (XAMPP Apache works the same
way).</p>
    <pre><code>cd department-site
python -m http.server 8000
# open http://localhost:8000/index.html</code></pre>
    <p>After each change, I saved the file and refreshed the browser (Ctrl + R). The localhost screenshot is in the report.</p>
  </section>

  <section id="hosting">
    <h2>Hosting Recommendation</h2>
    <p>I recommend <strong>shared hosting</strong>. The site is small and static, traffic is a few hundred visits a day, and shared hosting is the
cheapest option with the least administration. A VPS or dedicated server gives more control and scalability, but the department does not need
it yet.</p>
    <p>The <strong>domain name</strong> gives a readable address, <strong>DNS records</strong> (A record) point the name to the
server&amp;rsquo;s IP address, the <strong>hosting server</strong> stores and serves the files, and the <strong>control panel</strong> manages
files, SSL certificates and email.</p>
```

```
  </section>

  <section id="security">
    <h2>Security Considerations</h2>
    <ul>
      <li>Use HTTPS so form data is encrypted in transit.</li>
      <li>No passwords or secrets are placed in HTML comments or hidden inputs.</li>
      <li>Browser validation (required, pattern, min/max) helps users, but the server must validate again.</li>
      <li>Future database code must use parameterised queries to prevent SQL injection.</li>
      <li>Uploaded files must be checked for type and size and stored outside the web root.</li>
      <li>Any user input shown back on a page must be escaped to prevent XSS.</li>
      <li>The form collects only the data needed for the visit.</li>
      <li>The hosting account uses a strong password and two-factor authentication.</li>
    </ul>
  </section>

  <section id="sources">
    <h2>Sources</h2>
    <ul>
      <li>All images and the video were created by the student for this project.</li>
      <li><a href="https://developer.mozilla.org/en-US/docs/Web/HTML">MDN Web Docs: HTML reference</a></li>
    </ul>
  </section>
</main>

<footer>
```

```
  <p><small>&copy; <time datetime="2026">2026</time> Department of Robotics and Automation, SRM TRP Engineering College. Sample project for 2611CSSJ102 Web Essentials.</small></p>
  <p lang="ta">செயற்கை நுண்ணறிவு மற்றும் இயந்திரக் கற்றல் துறை</p>
 </footer>
</body>
</html>
```

style.css

```
/* Basic readability styles only (permitted by the assignment brief) */
body {
 margin: 0;
 font-family: Arial, Helvetica, sans-serif;
 line-height: 1.6;
 color: #1d2733;
 background: #f6f8fb;
}

header,
footer {
 background: #0e284a;
 color: #ffffff;
 padding: 1rem 1.5rem;
}

header p {
 margin: 0.2rem 0 0;
```

```
  color: #ffc43d;
}

nav ul {
  list-style: none;
  margin: 0.8rem 0 0;
  padding: 0;
  display: flex;
  flex-wrap: wrap;
  gap: 0.5rem;
}

nav a {
  color: #ffffff;
  padding: 0.3rem 0.7rem;
  border-radius: 4px;
}

nav a[aria-current="page"] {
  background: #ffc43d;
  color: #0e284a;
  font-weight: bold;
}

main {
  max-width: 960px;
  margin: 0 auto;
```

```
  padding: 1rem 1.5rem 2rem;
  background: #ffffff;
}

img,
video {
  max-width: 100%;
  height: auto;
}

figure {
  margin: 1rem 0;
}

figcaption {
  font-size: 0.9rem;
  color: #4a5a6e;
}

table {
  border-collapse: collapse;
  width: 100%;
  margin: 1rem 0;
}

th,
td {
```

```
  border: 1px solid #b8c4d2;
  padding: 0.4rem 0.6rem;
  text-align: left;
  vertical-align: top;
}

thead th {
  background: #e3ebf5;
}

caption {
  font-weight: bold;
  text-align: left;
  padding-bottom: 0.4rem;
}

fieldset {
  margin: 1rem 0;
  border: 1px solid #b8c4d2;
}

label {
  display: block;
  margin-top: 0.7rem;
  font-weight: bold;
}
```

```
.choice label {
  display: inline;
  font-weight: normal;
}

input,
select,
textarea {
  font: inherit;
  padding: 0.3rem;
}

.hint {
  font-size: 0.85rem;
  color: #4a5a6e;
}

pre {
  background: #eef2f7;
  padding: 0.8rem;
  overflow-x: auto;
}

footer a {
  color: #ffc43d;
}
```

```
/* Fix for test T12: keep wide tables readable on small screens */
.table-wrap {
  overflow-x: auto;
}

@media (max-width: 600px) {
  th,
  td {
    padding: 0.25rem 0.35rem;
    font-size: 0.85rem;
  }
}
```

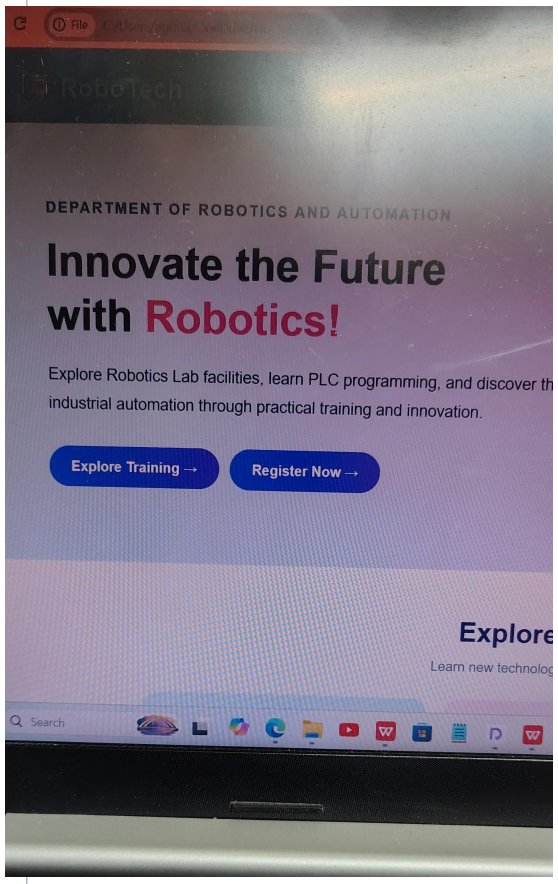


Figure 1: Home page (index.html)

Figure 2: About page (about.html)

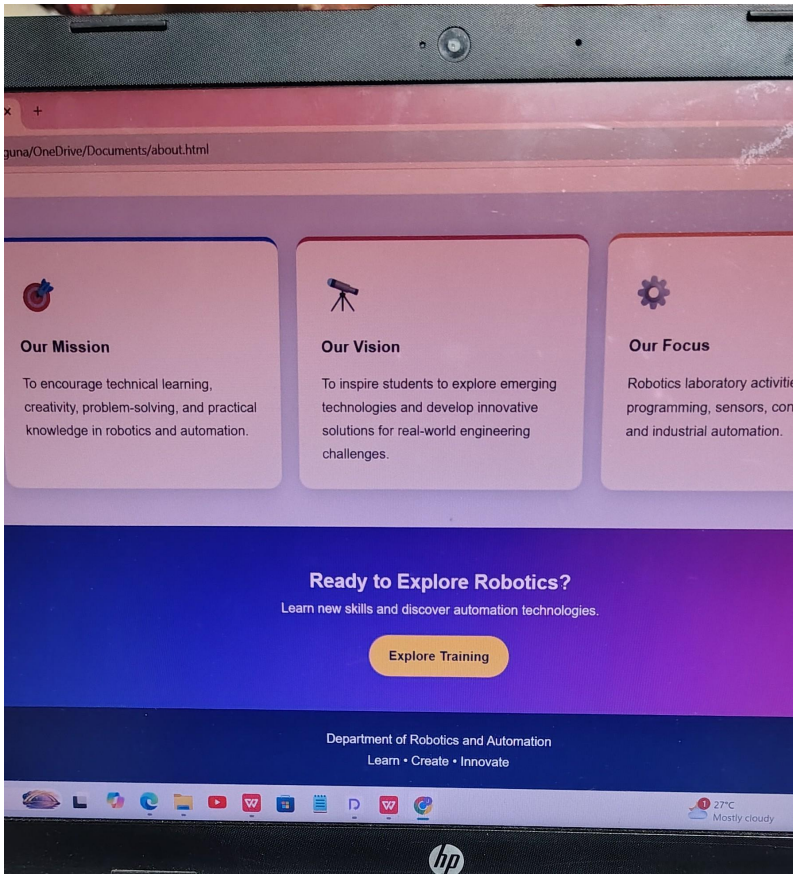


Figure 3: Courses page (courses.html) with the accessible table

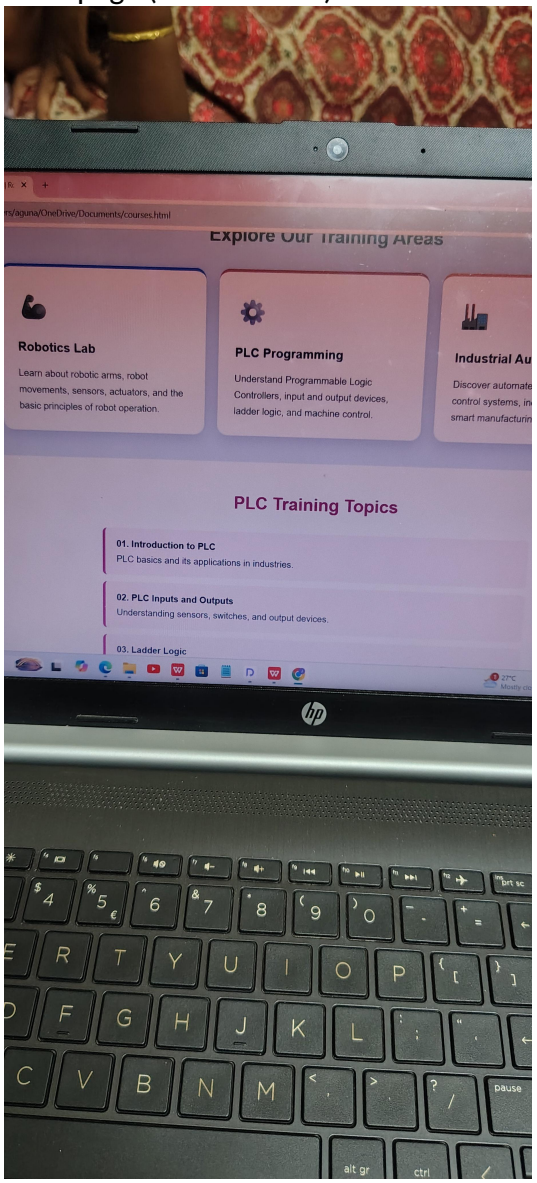
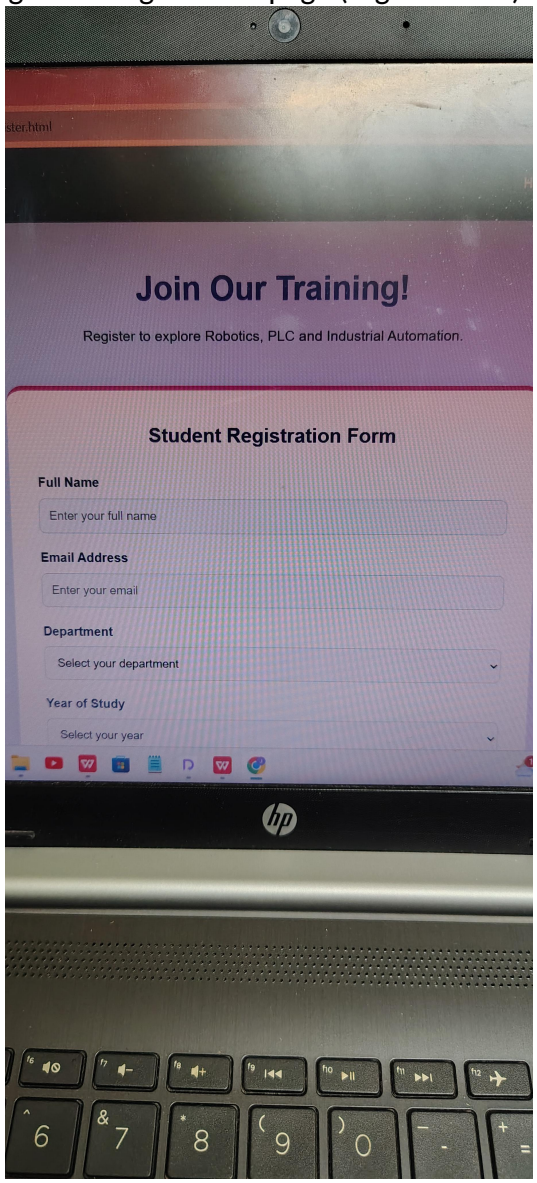


Figure 4: Registration page (register.html)



5. Testing Record

Test ID	Test	Expected result	Actual result	Status
T01	Open index.html through localhost	Home page loads with status 200	Returned 200; page displayed	Pass
T02	Use navigation from every page	Every destination opens correctly	36 links checked, all 200; current page marked with aria-current	Pass
T03	Activate fragment link	Browser moves to matching id	URL became contact.html#office-hours; section scrolled into view	Pass
T04	Load images and media	No missing-file errors	Video did not play in one browser (MP4 only). Fixed by adding a WebM source and poster.	Fixed – Pass
T05	Navigate by keyboard	Logical focus order	Nav links → name → email → phone → date → visitors → mode → interests → session → question → file → Submit	Pass
T06	Submit empty form	Required-field feedback appears	"Please fill out this field." shown on Full name	Pass
T07	Enter invalid email / number	Browser reports invalid value	Email without @, visitors = 9 and phone 12345 rejected	Pass
T08	Inspect labels	Every control has a label	13 controls, each with one label and a name	Pass
T09	Inspect table headings	Relationships are clear	Caption, 6 column and 5 row headings with scope; rowspan and colspan correct	Pass
T10	Validate HTML	No structural errors	W3C Nu HTML Checker: 0 errors, 0 warnings on all six pages	Pass
T11	Inspect Network panel	No 404 responses	All responses 200, 206 or 304	Pass
T12	Resize browser window	Content remains readable	Course table overflowed at 375 px. Fixed with a scrollable wrapper and smaller cell padding.	Fixed – Pass

6. Result

The multi-page HTML5 website for the Department of Robotics and Automation was planned, created and tested successfully. All six compulsory pages were served through the local Web server at http://localhost:8000, the navigation, table, registration form and media worked as expected, and the W3C validator reported no errors. All twelve tests passed after the two defects found during testing were corrected. Hence the aim of the assignment was achieved, and the output was verified.

Robotics and Automation Website – Image Assets

Student Name: Bavadharani L Register Number: TRP26CS34

Figure 1: Robotics Laboratory

This image represents an industrial robotic arm, conveyor system and automation workspace used in the website Home and About pages.

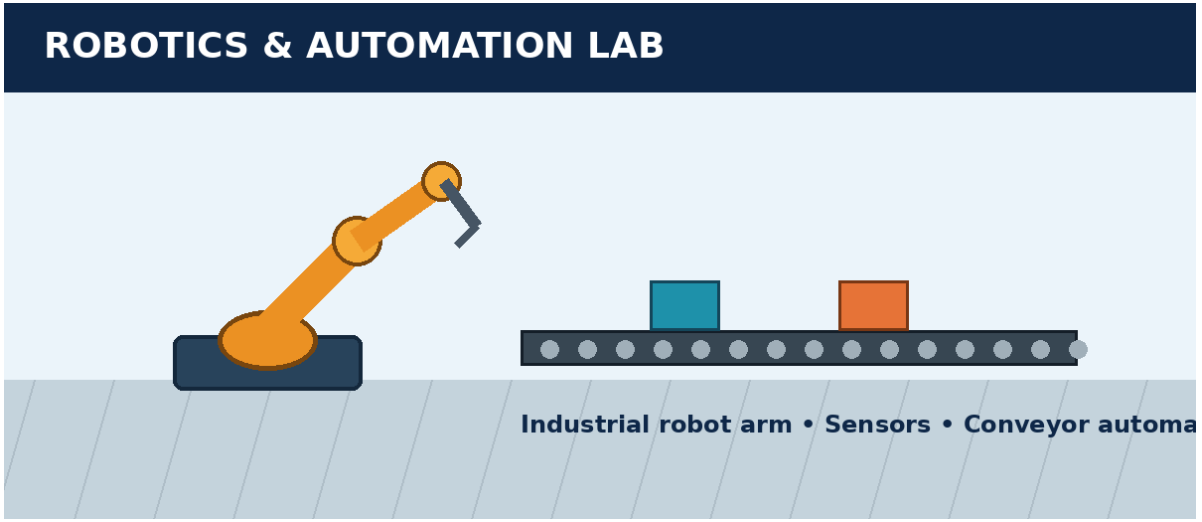
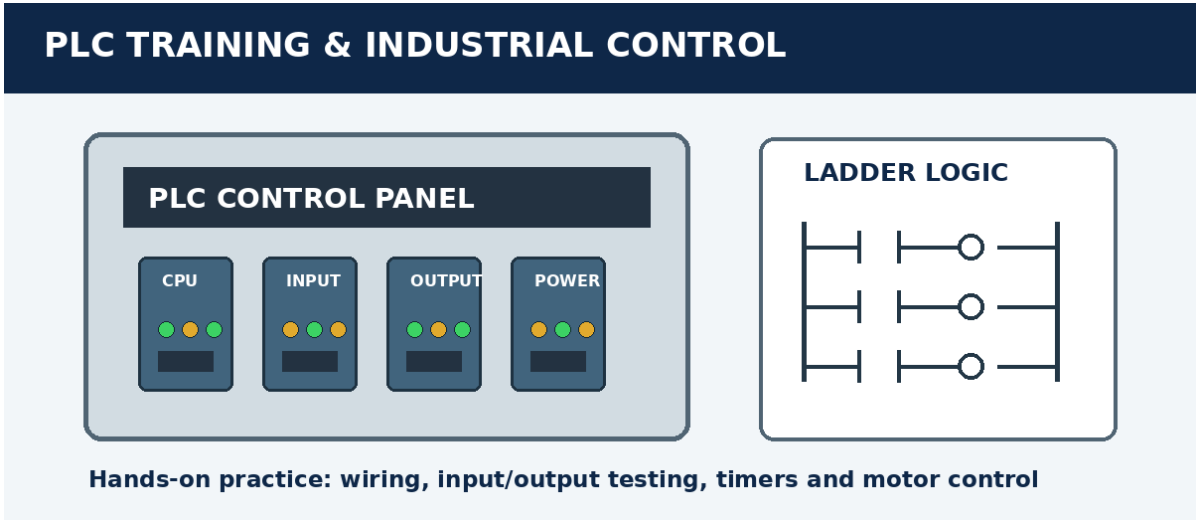


Figure 2: PLC Training Laboratory

This image represents a PLC control panel with CPU, input/output modules and a ladder-logic training display.



Website image tags

```
<figure>
  <img src="images/robotics-lab.png" alt="Industrial robotic arm and conveyor automation setup" width="1200" height="520">
  <figcaption>Figure 1: Robotics and Automation Laboratory.</figcaption>
</figure>
<figure>
  <img src="images/plc-training.png" alt="PLC control panel with CPU, input and output modules and ladder logic" width="1200"
height="520">
  <figcaption>Figure 2: PLC Training Laboratory.</figcaption>
</figure>
```

Website CSS tags and styling

```
body { font-family: Arial, sans-serif; margin: 0; background: #f2f6f9; color: #172b3a; }
header { background: #0e2748; color: white; padding: 24px; text-align: center; }
```

```
nav a { color: white; padding: 10px 14px; text-decoration: none; }
main { max-width: 1100px; margin: auto; padding: 24px; }
figure img { max-width: 100%; height: auto; border-radius: 12px; }
section { background: white; margin: 18px 0; padding: 20px; border-radius: 10px; }
footer { background: #0e2748; color: white; padding: 16px; text-align: center; }
```